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Simplifying Roots

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Simplifying Roots

A **radical** is an expression that contains the symbol $\sqrt{}$. You already know how to simplify some radicals.

Example 1

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$\sqrt{81} = 9$ because $9 \times 9 = 9^2 = 81$.

You can find the square root of 81 because 81 is a **perfect square**. A perfect square is formed when you raise a number to the second power. The first 12 perfect squares are 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, and 144.

Most numbers are not perfect squares. Without a calculator, you may not be able to find the exact value, but you can estimate the value.

Example 2

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Estimate the value of $\sqrt{70}$.

$\sqrt{70}$ is between $\sqrt{64}$ and $\sqrt{81}$. You know that $\sqrt{64} = 8$ and $\sqrt{81} = 9$, so you can conclude that the value of $\sqrt{70}$ is **between 8 and 9**.

As you continue learning about algebra, you will need to be able to write a radical in simplest form. You should memorize the properties in the box below. The properties are true only for numbers that are greater than or equal to zero.

Properties of Square Roots

$a \geq 0$ and $b \geq 0$

Product property $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

Quotient property $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

Example 3

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Write $\sqrt{54}$ in simplest form.

Step 1

Use the product property. Factor the number inside. Look for a perfect square factor.

$\sqrt{54} = \sqrt{9} \times \sqrt{6}$

Step 2

Find the square root of the perfect square: $\sqrt{9} = 3$

Write it outside the bracket. $3\sqrt{6}$ means $3 \times \sqrt{6}$

$3\sqrt{6}$

The expressions $\sqrt{54}$ and $3\sqrt{6}$ have equal value.

Example 4

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Write $\sqrt{\frac{7}{25}}$ in simplest form.

Step 1

Use the quotient property. Write the quotient using two radicals.

$\sqrt{\frac{7}{25}} = \frac{\sqrt{7}}{\sqrt{25}}$

Step 2

Find the square root of the perfect square: $\sqrt{25} = 5$

$\frac{\sqrt{7}}{5}$ or $\frac{1}{5}\sqrt{7}$

The expressions $\sqrt{\frac{7}{25}}$ and $\frac{1}{5}\sqrt{7}$ have equal value.

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Lesson 1: Exponents and Roots

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